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Anxiety, Stress, & Coping
An International Journal

ISSN: 1061-5806 (Print) 1477-2205 (Online) Journal homepage: www.tandfonline.com/journals/gasc20

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To cite this article: Yi Ling Tan, Vina Yang Xiu Chang, Wei How Darryl Ang, Wen Wei Ang & Ying Lau (2025) Virtual reality exposure therapy for social anxiety disorders: a meta-analysis and meta-regression of randomized controlled trials, Anxiety, Stress, & Coping, 38:2, 141-160, DOI: 10.1080/10615806.2024.2392195

To link to this article: https://doi.org/10.1080/10615806.2024.2392195

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ANXIETY, STRESS, & COPING
2025, VOL. 38, NO. 2, 141–160
https://doi.org/10.1080/10615806.2024.2392195

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Virtual reality exposure therapy for social anxiety disorders: a meta-analysis and meta-regression of randomized controlled trials

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ABSTRACT

Background and objectives: Virtual reality exposure therapy offers a unique opportunity to treat social anxiety disorder. This meta-analysis aims to evaluate the effectiveness of virtual reality exposure therapy compared to waitlist comparators or other interventions for individuals with social anxiety disorder in alleviating anxiety symptoms.

Methods: A three-step comprehensive search for the randomized controlled trials of virtual reality exposure therapy was conducted from inception to 7 December 2023. The overall effect was measured using Hedges' *g* and determined using *t*-statistics at a significance level of $p < 0.05$. Sensitivity, subgroup, and meta-regression analyses were carried out.

Results: A total of 17 randomized control trials were retrieved from nine electronic databases. Virtual reality exposure therapy has greater efficacy than waitlist comparators in reducing anxiety symptoms at post-intervention and follow-up assessment. Virtual reality exposure therapy demonstrates a similar effect to other interventions at post-intervention and follow-up assessment. We observed a greater effect for participants with symptomatic social anxiety when we combined the intervention with cognitive behavioral therapy compared to its counterpart. Meta-regression analyses found no significant covariate.

Conclusions: Overall, virtual reality exposure therapy can provide supplementary therapy for improving anxiety symptoms. Additional high-quality and large-scale trials with long-term follow-up are needed.

ARTICLE HISTORY
Received 25 April 2023
Revised 25 June 2024
Accepted 9 August 2024

KEYWORDS
Virtual reality exposure therapy; social anxiety disorders; anxiety symptoms; systematic review; meta-analysis; meta-regression

Introduction

Social anxiety disorder (SAD), also known as social phobia, is the third most common psychological disorder, with a lifetime prevalence ranging from 8% to 36% (Jefferies & Ungar, 2020; McEnery et al., 2019) owing to different diagnostic thresholds. SAD is characterized by excessive fear and discomfort

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Supplemental data for this article can be accessed online at https://10.1080/10615806.2024.2392195.
This article has been corrected with minor changes. These changes do not impact the academic content of the article.

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in social situations where the individual is exposed to strangers in social settings (American Psychiatric Association & Association, 2013). Individuals with SAD have a heightened fear of interaction with others and worry about receiving scrutiny (Hofmann & DiBartolo, 2014). The common triggers related to fear are speaking up in public, being the center of attention, or talking to strangers (Hofmann & DiBartolo, 2014). Symptoms of SAD significantly impact an individual's daily life, relationships, and work, resulting in lowered productivity, social impairment, and poor quality of life (Dryman et al., 2016; National Collaborating Centre for Mental, 2015; Stein et al., 2017).

SAD has a debilitating nature but is often unrecognized and untreated (National Collaborating Centre for Mental, 2015). Most SAD patients display a tendency to hide their problems (Bandelow & Michaelis, 2015) and withhold treatment unless symptoms are persistent, pervasive, or accompanied by other critical psychiatric conditions (Lin et al., 2019). Individuals with SAD are vulnerable to stigma-related concerns, deterrents to seeking help (Clement et al., 2015; National Collaborating Centre for Mental, 2015) and a high non-compliance rate (Wise, 2017). In vivo exposure therapy is a gold-standard treatment that aims to reduce the fear associated with a trigger (Scheurich et al., 2019). Individuals confront themselves repetitively in real life with stimuli over a long period until fear decreases (Scheurich et al., 2019). However, some limitations inherent to in vivo delivery are observed, such as patient aversion, difficulty in arranging sessions, difficulty in imagining or visualization of situations, and impractical reconstruction of scenarios (Reeves, Curran, et al., 2022; Scheurich et al., 2019). As technology continues to improve and shape health management, a new technology-based form of exposure therapy can be an alternative.

Virtual reality exposure therapy

Virtual reality exposure therapy (VRET) uses computer-generated virtual environments, and individuals with SAD can be systematically exposed to feared stimuli in a highly controlled environment. Thus, each exposure can be flexibly fine-tuned to individual needs (Wilhelm et al., 2005). In VRET, individuals fully immerse into virtual environments using display systems, head-mounted displays (HMD), and synthesized sounds (Boeldt et al., 2019). These devices help stimulate multiple senses, thereby improving the sense of realism and enhancing the memory of the virtual environment (Maples-Keller et al., 2017). Sessions can be monitored, and an area of concern can be isolated. The secured environment allows individuals to express thoughts and emotions that may be suppressed in vivo (Carvalho et al., 2010). VRET is an individualized, gradual, precisely controlled, and immersive exposure treatment that can be easily implemented by therapists (Boeldt et al., 2019). The therapist helps the individuals emotionally engage through the process, discouraging the use of safety behaviors or coping strategies (Maples-Keller et al., 2017). This interaction potentially improves the therapist-patient alliance (Carvalho et al., 2010), which is important in ensuring positive therapeutic outcomes (Ardito & Rabellino, 2011).

VRET is a valuable addition to exposure therapy for reducing anxiety symptoms (Maples-Keller et al., 2017). Evidence shows that individuals prefer VRET over in vivo exposure therapy (Garcia-Palacios et al., 2001; Garcia-Palacios et al., 2007). Although one review (Powers & Emmelkamp, 2008) has shown the superiority of VRET over in vivo exposure among participants with mixtures of specific phobia, social phobia, post-traumatic stress disorders, and panic disorders, another review (Wechsler et al., 2019) has found in vivo exposure to be superior to VRET in social phobia. Two other reviews (Chesham et al., 2018; Horigome et al., 2020) claim no difference in effect between VRET and other standard interventions for social anxiety. Given these divergent findings in previous literature, additional investigations are needed.

Mechanisms of virtual reality exposure therapy on anxiety symptoms

Individuals with SAD have pathological fear structures, and they do not accurately represent reality because they involve excessive response elements and resistance to modification (Rauch & Foa,

2006). Emotional Processing Theory suggested that the fear structures of SAD included dysfunctional cognition, so modification of the pathological elements of the fear structure was crucial via exposure therapy (Rupp et al., 2017). The possible mechanism of VRET in improving anxiety symptoms is based on Emotional Processing Theory (Foa et al., 2006; McGuire et al., 2014). Emotional processing occurs under two important conditions for modification of a fear structure: (1) activation of the fear structure, and (2) incorporation of new information and replacement of pathological elements with realistic ones (Foa et al., 2006; Rauch & Foa, 2006).

The theory suggests modifying the fear structure because pathological fear structures cause heightened self-focused attention and negative affect, causing anxiety symptoms (Foa et al., 2006; Suárez et al., 2008). Exposure therapy contains a repeating and systematic confrontation with imaginal exposures (Rauch & Foa, 2006). Interference with the pathological fear structure by contradicting existing elements is key to recovery (Foa et al., 2006). Individuals with SADs confront a feared stimulus in a planned, organized, and gradual manner (McGuire et al., 2014). It is replaced with a non-fear stimulus when corrective information is incompatible with this feared stimulus (Foa et al., 2006).

VRET provides a tailored environment to fear structures for the patients and real-time feedback (Bohil et al., 2011; Lindner et al., 2017). This allows therapists to control exposure to feared situations, enabling individuals to confront their fears gradually (Maples-Keller et al., 2017). For instance, patients who are afraid of public speaking may realize that the audience is not as critical as they imagined or that making mistakes is a normal part of public speaking. One meta-analysis revealed that within-session and between-session habituation were positively related to treatment outcomes in exposure therapy (Rupp et al., 2017). Virtual reality (VR) technology integrates computer graphics, visual displays, motion tracking, and another sensory device to provide the participant with a multisensory and realistic experience (Powers & Emmelkamp, 2008). As sessions progress, the complexity of the interaction between individuals and fear stimuli is increased through cognitive modification, habituation, and distraction, thereby lowering anxiety symptoms (McGuire et al., 2014).

Literature gaps

A growing number of systematic reviews have investigated the effects of technology-assisted intervention (Kampmann et al., 2016b) or VRET (Carl et al., 2019; Chesham et al., 2018; Horigome et al., 2020; Powers & Emmelkamp, 2008; Wechsler et al., 2019) on various outcomes (Oing & Prescott, 2018; Qian et al., 2020). However, these reviews have some limitations, such as using few databases (Carl et al., 2019; Kampmann et al., 2016b; Oing & Prescott, 2018; Powers & Emmelkamp, 2008), discussing a mixture of psychotherapy (Kampmann et al., 2016b; Qian et al., 2020), targeting different psychological disorders (Oing & Prescott, 2018; Qian et al., 2020; Wechsler et al., 2019), lacking quality assessment (Chesham et al., 2018), combining different types of research design (Horigome et al., 2020; Oing & Prescott, 2018; Qian et al., 2020), using a mixture of control condition and standard treatment as a comparator (Kampmann et al., 2016b). Regarding the type of comparator, if the comparator of VRET comprises control conditions and other interventions in a meta-analysis, it makes the results difficult to interpret. Hence, we compared the effects of VRET for two comparisons in separate meta-analyses.

Further, reviews also reported inconsistent findings (Horigome et al., 2020; Powers & Emmelkamp, 2008), adopting a small number of randomized control trials (RCTs) (Chesham et al., 2018; Wechsler et al., 2019), and using solely narrative analysis (Qian et al., 2020). One literature review has focused on current methods and important aspects of VRET for SAD (Emmelkamp et al., 2020), and a few have focused on the long-term efficacy of VRET (Horigome et al., 2020). Few performed meta-regression to explore the potential covariates, and none reported the certainty of evidence. Given that research on VR exposure therapy is a fast-growing field, previous reviews were conducted for over three years. Up-to-date evidence is required.

Given the gaps mentioned above, the current review has the following two hypotheses: (1) VRET demonstrates a greater effect than the control condition (waitlist comparators) on anxiety symptoms at post-intervention and follow-up assessment; (2) VRET demonstrates a similar effect to other interventions (in vivo exposure therapy or other psychotherapy). We also compare the effect sizes in a series of subgroup analyses of two comparisons according to region, nature of population, and co-intervention. We further explore any potential covariates on treatment effects in both comparisons.

Methods

This review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist (Page et al., 2021) in Table S1. The review was prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO) with registration number: CRD42021239015.

Search strategies

A scoping search was conducted to identify similar reviews in the PROSPERO databases, systematic review journals, Cochrane databases of systematic review, and Joanna Briggs Institute databases to prevent the publication of a similar review. Subsequently, the search for studies was conducted according to the three-step search strategy (Higgins et al., 2019). First, a search was conducted across nine electronic databases: PubMed, Embase, Cochrane, Cumulated Index to Nursing and Allied Health Literature, Institute of Electrical and Electronics Engineers Xplore, PsycINFO, Scopus, Web of Science, and ProQuest dissertations and theses global from inception to 7 December 2023. The search strategy is detailed in Table S2. No limits were imposed on the publication date and time. Next, we searched clinical trial registries, specifically ClinicalTrials.gov and the Cochrane Controlled Register of Trials. An email was sent to trial authors to gather preliminary data for data analysis. A final review and hand-searching through reference lists of other reviews and grey literature such as Google Scholar and CogPrints were also performed to determine if any study was missed.

Eligibility criteria

The eligibility criteria are detailed in Table S3. The population targeted were individuals diagnosed with SAD, public-speaking anxiety, or social anxiety symptoms according to diagnostic criteria (American Psychiatric Association & Association, 2013). We also included participants who presented with elevated social anxiety symptoms, as determined by the respective cut-off scores of the following scales: Liebowitz Social Anxiety Scale (LSAS) (Mennin et al., 2002) with a cut-off of ≥ 31 (Lin et al., 2019) and a cut-off of ≥ 20 (Nazligul et al., 2019), Personal Report of Confidence as Speaker (PRCS) inventory (Paul, 1967) with a cut-off of ≥ 16 (Harris et al., 2002), Public-Speaking Anxiety Scale (PSAS) (Bartholomay & Houlihan, 2016) with a cut-off of ≥ 55 (Kahlon, Gjestad, et al., 2023; Kahlon, Lindner, et al., 2023) and a cut-off of ≥ 60 (Lin et al., 2019; Lindner et al., 2019; Reeves, Elliott, et al., 2021) or self-diagnosis without mention about a cut-off point (Safir et al., 2012).

The intervention was exposure therapy incorporating VR technology using immersive systems (e.g., HMD) and head tracking. VRET can stand alone or be combined with cognitive-behavior therapy (CBT), cognitive therapy (CT) or exposure program (EXP). To examine the two hypotheses of this review, two comparators were used. The first comparator was the waitlist comparator, and the second comparator was other interventions. The outcome was anxiety symptoms measured by Brief Fear of Negative Evaluation (FNE-B) (Liu & Lowe, 2016), LSAS (Rytwinski et al., 2009), PSAS (Bartholomay & Houlihan, 2016), or Social Intervention Anxiety Scale (SIAS) (Mattick & Clarke, 1998). The study design of the articles included all types of RCTs. Article selection, data extraction,

and quality assessment between two independent reviewers were calculated using Cohen's kappa (*k*) statistics (McHugh, 2012). Values of *k* were interpreted as follows: ≤ 0, no agreement; 0.01–0.20, none to slight; 0.21–0.40, fair; 0.41–0.60, moderate; 0.61–0.80, substantial; and 0.81–1.00, almost perfect agreement (McHugh, 2012).

Study selection and management

All studies generated from the search were imported into Endnote X20 (The EndNote Team, 2022). Following the automated and manual removal of duplicates, two reviewers (YL and DA) screened all studies by their titles and abstracts. Articles were selected based on the eligibility criteria. When disagreements occurred, we brought in a third reviewer (YL) to resolve disagreements.

Data extraction

We collected data using a standardized data extraction form based on the Cochrane Handbook for Systematic Reviews of Intervention (Higgins et al., 2019). The data collection method was piloted to facilitate the completeness of the data collection form, focusing on data needed for meta-analysis to ensure usability. Independent extraction of data by different reviewers (YLT and DA) was conducted to reduce errors (Li et al., 2023). Items extracted from the characteristics of RCTs included authors, year, country, design, population characteristics, criteria, age, sample size, intervention, comparator, outcome, measures, attrition rate, intention-to-treat (ITT) analysis, missing data management, protocol, registration, and grant support. Items for intervention included approach, VR system, intervention content, co-intervention, regime (session, length, frequency, and duration), and follow-up. We contacted authors for additional information when data from published and unpublished trials were missing.

Risk of bias version 2

The Cochrane risk of bias tool version 2 (Sterne et al., 2019) was used by two reviewers (YLT and DA) to independently assess the quality of the individual studies. The value of *k* was 0.86, indicating substantial inter-rater agreement. A third author (YL) was involved in resolving any difference through discussions. The risk of bias was assessed based on several domains: (1) the randomization process; (2) deviations from the intended intervention; (3) missing outcome data; (4) measurement of the outcome; and (5) selection of the reported result. In each domain, two reviewers signaled questions to be answered with "yes," "probably yes," "probably no," "no," or "no information." The risk of bias is categorized as "low," "high," or "some concerns" by the RoB 2.0 algorithmic tool (Sterne et al., 2019).

Data analysis

Meta package (Schwarzer & Schwarzer, 2012) in R software was used to conduct the meta-analysis, subgroup analyses, and meta-regression. Continuous outcomes were measured using the mean and standard deviation with 95% confidence intervals (CIs). Considering the small sample size in selected RCTs, Hedges's *g* was used to compare the effect sizes between VRET and two comparators because of its precision for studies with small sample sizes (Brydges, 2019). For each trial, we calculated the changes in the mean and standard deviation of the pre-test and post-test as the pre-post-test Hedges' *g* effect size. Independent samples served as the unit of analysis to address any potential issues relating to effect size dependency (Gucciardi et al., 2021). When studies utilized two instruments to measure the same outcome, we included only one measure and chose the widely validated and common scale among the included studies.

The restricted maximum likelihood approach was used as the estimator for random-effect meta-analysis (Jamshidian, 2004), guided by the Hartung-Knapp-Sidik-Jonkman (HKSJ) method (IntHout

et al., 2014). The inverse-variance method was used, and the overall effect was computed using *t*-statistics at a significance level of $p < 0.05$ (Deeks et al., 2020; Hedges & Salkind, 2010). Heterogeneity was assessed using the I^2 statistic, where values between 30%–60%, 50%–90%, and 75%–100% were considered moderate, substantial, and considerable levels of heterogeneity, respectively (Aromataris & Munn, 2020).

Subgroup analyses involving the region (Asia, Europe vs. North America), nature of population (diagnostic SAD vs. symptomatic SAD), and presence of co-intervention (solely VRET vs. VRET plus other intervention) were also performed to address potential heterogeneity (Rothwell, 2005). The test of subgroup differences was evaluated using the *Q* statistic (χ^2). A subgroup difference was ascertained when the *p*-value was less than 0.1 (Richardson et al., 2019). A meta-regression was conducted to examine the effect of potential covariates (publication year, number of sessions, attrition rate, and sample size) on anxiety symptoms. The coefficient β, which represents a change in the dependent variable relative to the unit change in the covariates, was used to express the relationships (Higgins et al., 2019). The χ^2 value and $p < 0.05$ were used to examine the treatment effect (Higgins et al., 2019). A narrative synthesis was performed if the studies did not provide sufficient information for a meta-analysis.

Certainty of evidence

The assessment of certainty of evidence across outcomes was conducted independently by two reviewers (LYT and DA) according to the Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) criteria (Guyatt et al., 2008). The following indicators were used to determine the certainty of information: (1) methodological limitation; (2) inconsistency; (3) indirectness; (4) imprecision; and (5) publication bias. The eventual rating was classified as very low, low, moderate, or high based on the justifications to support the ratings (Guyatt et al., 2011). Publication bias was determined using the Egger regression test and visual inspection of the funnel plot (Sterne et al., 2011). An asymmetry in the funnel plot and a $p < 0.05$ based on the Egger's test (Sterne et al., 2011) are indications of publication bias.

Results

The PRISMA flowchart (Figure 1) depicts the selection process for the studies. An initial search of nine databases and two main clinical registries identifies 10,643 records. After using Endnote to remove duplicates, 7,099 records are screened, and 6,996 articles are excluded based on irrelevant titles and abstracts. We selected 103 full-text articles for eligibility assessment; we excluded 79 articles were excluded (Table S4) and 17 ongoing trials were excluded (Table S5) with reasons. Finally, 17 RCTs in 20 publications are selected (Anderson et al., 2013; Anderson et al., 2017; Bouchard et al., 2017; Burton et al., 2013; Harris et al., 2002; Kahlon, Gjestad, et al., 2023; Kahlon, Lindner, et al., 2023; Kampmann et al., 2016a; Kim et al., 2022; Lin et al., 2019; Lindner et al., 2019; Moldovan & David, 2014; Nazligul et al., 2019; Reeves, Elliott, et al., 2021; Safir et al., 2012; Sarpourian et al., 2022; Stefaniak et al., 2022; Wallach et al., 2009; Wallach et al., 2011; Zainal et al., 2021). The results of article selection show that $k = 0.74$, indicating substantial inter-rater agreement (McHugh, 2012).

Study characteristics

Table 1 summarizes the characteristics of 17 RCTs in 943 participants from 2002 (Harris et al., 2002) to 2023 (Kahlon, Gjestad, et al., 2023). The RCTs are conducted across twelve countries, namely, USA ($n = 5$), Canada ($n = 1$) (Bouchard et al., 2017), Iran ($n = 2$) (Sarpourian et al., 2022), Israel ($n = 2$) (Safir et al., 2012; Wallach et al., 2011), the Netherlands ($n = 1$) (Kampmann, et al., 2016a), Norway ($n = 1$) (Kahlon, Gjestad, et al., 2023; Kahlon, Lindner, et al., 2023), Poland ($n = 1$) (Stefaniak et al., 2022), Singapore ($n = 1$) (Lin et al., 2019), Sweden ($n = 1$) (Lindner et al., 2019), South Korea ($n = 1$)

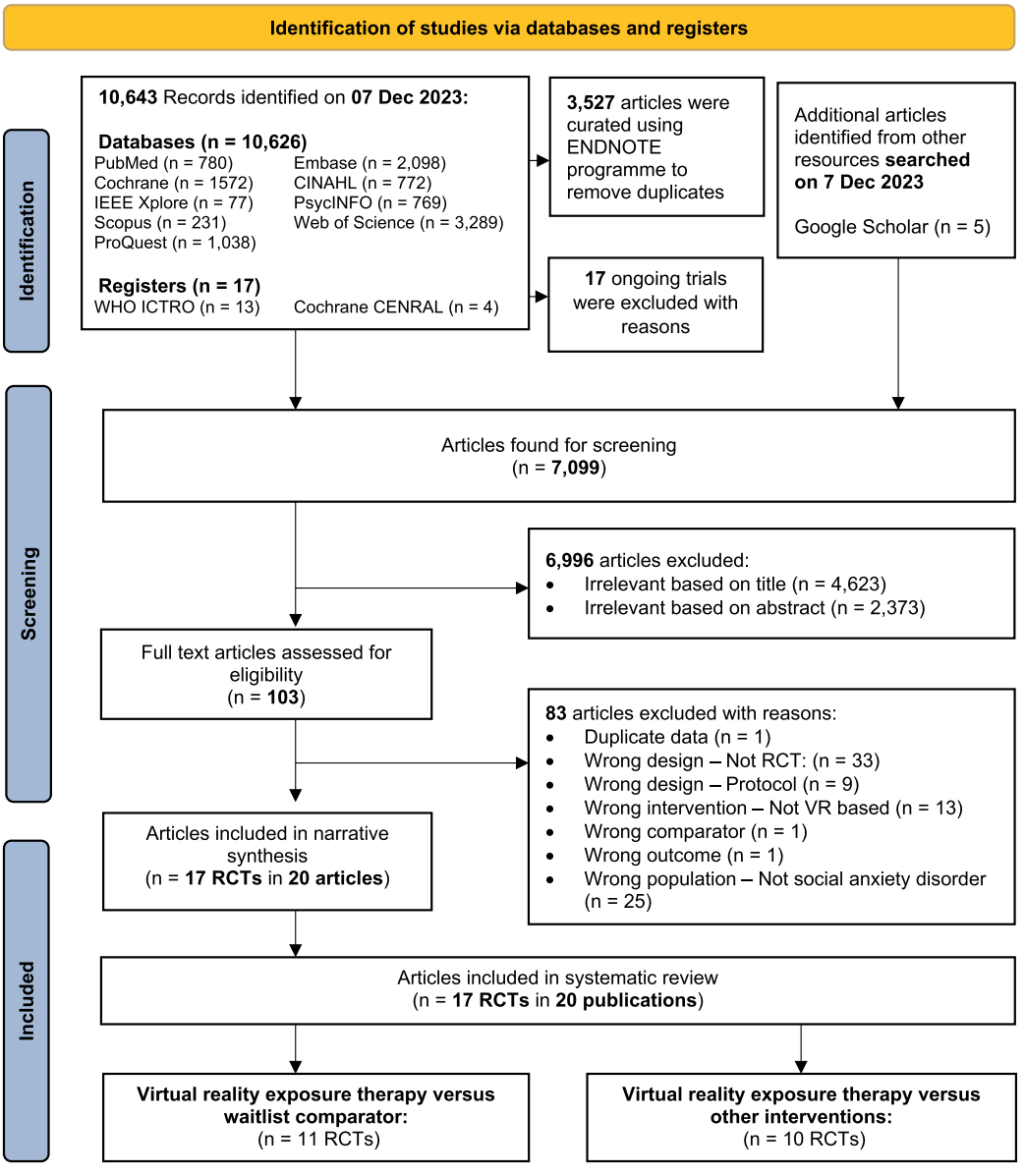


Figure 1. PRISMA flowchart. Note *n* = Number of records; RCT = Randomized Controlled Trial.

(Kim et al., 2022), Turkey (*n* = 1) (Nazligul et al., 2019), and United Kingdom (*n* = 1) (Reeves, Elliott, et al., 2021). Sample sizes of the studies range from 14 (Nazligul et al., 2019) to 112 participants (Wallach et al., 2009).

A detailed description of VRET for the 17 trials is presented in Table S6. Nine trials use solely VRET, eight trials use a combination of VRET with CBT, CT or EXP as the intervention. VR facilities include a virtual conference room, classroom, and auditorium using various types of HMD and potable headbands. The number of sessions ranges from 1 (Lindner et al., 2019) to 14 (Bouchard et al., 2017) and is held either multiple times per week (Zainal et al., 2021) or once a week (Bouchard et al., 2017). The length of each session ranges from 8 min (Nazligul et al., 2019) to 60 min (Lin et al., 2019; Wallach

Table 1. Characteristics of 17 randomized controlled trials in 20 articles.

Author, year (country)	Design	Participant characteristics (criteria)	Age (M ± SD)	Sample size	Intervention	Comparator(s)	Outcome (measure)	Attrition rate	ITT/ MDM	Protocol/ registration/ grant
Anderson et al., 2013, 2017 (USA)	3-arm RCT	Social anxiety disorder (DSM-4-TR)	42 ± 13.17	T: 97 I: 30 $C_{1,2}$: 28,39	VRET	C_1: Waitlist C_2: In-vivoET	Anxiety symptoms (FNE-B)	22.68%	N/N	N/N/Y
Bouchard et al., 2017 (Canada)	3-arm RCT	Social anxiety disorder (DSM-5)	34.5 ± 11.9	T: 59 I: 17 $C_{1,2}$: 20,22	VRET + CBT	C_1: Waitlist C_2: In-vivoET + CBT	Anxiety symptoms (LSAS-SR)	16.9%	N/N	Y/Y/Y
Burton et al., 2013 (USA)	3-arm RCT	Social anxiety disorder (DSM-4)	40 ± 12	T: 65 I: 32 C_1: 33	VRET + CBT	C_1: In-vivoET + CBT	Anxiety symptoms (FNE-B)	23.6%	N/Y	N/N/Y
Harris et al., 2002 (USA)	2-arm RCT	Public speaking anxiety (PRCS > 16)	NR	T: 16 I: 8 C_1: 8	VRET	C_1: Waitlist	Anxiety symptoms (LSAS)	12.5%	N/N	N/N/N
Kahlon, Gjestad, et al., 2023; Kahlon, Lindner, et al., 2023 (Norway)	4-arm RCT	Public speaking anxiety (PSAS ≥ 55)	14.16 ± 0.99	T: 100 $I_{1,2}$: 20, 20 $C_{1,2}$:40,20	I_1: VRET + NA I_2: VRET + online EXP	C_1: Waitlist + online PE C_2: Online PE + online EXP	Public speaking anxiety symptoms (PSAS)	27%	Y/N	N/Y/Y
Kampmann et al, 2016a (Netherlands)	3-arm RCT	Social anxiety disorder (DSM-4)	36.9 ± NR	T: 60 I: 20 $C_{1,2}$: 20,20	VRET	C_1: Waitlist C_2: In-vivoET	Anxiety symptoms (LSAS-SR)	16.7%	Y/N	N/Y/Y
Kim et al., 2022 (South Korea)	2-arm RCT	Social anxiety disorder (DSM-5)	19–30	T: 52 I: 24 C_1: 28	VRET	C_1: Waitlist	Anxiety symptoms (LSAS)	21.2%	N/N	N/N/Y
Lin et al., 2019 (Singapore)	2-arm RCT	Social anxiety symptoms (LSAS ≥ 31, PSAS ≥ 60)	25.6 ± 3.96	T: 50 I: 25 C_1: 25	VRET	C_1: Waitlist	Anxiety symptoms (LSAS)	12%	Y/Y	N/Y/Y
Lindner et al., 2019 (Sweden)	2-arm RCT	Public speaking anxiety (PSAS ≥ 60)	31.36 ± NR	T: 50 I: 25 $C_{1,2}$: 25	VRET + CBT	C_1: Waitlist C_2: In-vivo transition program	Anxiety symptoms (LSAS)	12%	Y/Y	N/Y/N
Moldovan & David, 2014 (USA)	2-arm RCT	Social phobia, flight phobia, or acrophobia (DSM-4)	>18	T: 32 I: 16 C_1: 16	VRET + CBT	C_1: Waitlist	Anxiety symptoms (LSAS)	0%	N/N	N/N/N

Nazligul et al., 2019 (Turkey)	2-arm RCT	Public speaking anxiety symptoms (LSAS $\geq$ 20)	21.36 $\pm$ 1.08	T: 14 I: 7 C_1: 7	VRET	C_1: CBT-based psychoeducation	Anxiety symptoms (LSAS)	0%	N/N	N/N/N
Reeves, Elliott et al, 2021 (UK)	3-arm RCT	Public speaking anxiety (PSAS $\geq$ 60)	26.0 $\pm$ 7.53	T: 51 I_1: 17 I_2: 16 C:18	I_1: VRET (360∘Audience) I_2: VRET (360∘Empty)	C_1: Waitlist	Public speaking anxiety symptoms (PSAS)	7.8%	N/N	N/Y/N
Sarpourian et al., 2022 (Iran)	2-arm RCT	Public speaking anxiety (Self-diagnosis)	21–50	T: 30 I: 15 C_1:15	VRET	C_1: Counseling	Anxiety symptoms (LSAS)	0%	N/N	N/N/Y
Stefaniak et al., 2022 (Poland)	3-arm RCT	Social anxiety disorder (DSM-4-TR)	31.6 $\pm$ NR 18–50	T: 91 I_1: 30 I_2: 31 C_1:30	I_1: VRET I_2: VRET + CBT	C_1: CBT	Anxiety symptoms (LSAS)	19.8%	N/Y	N/Y/Y
Wallach et al., 2009; Safir et al., 2012 (Israel)	3-arm RCT	Public speaking anxiety (DSM-4-TR)	27.49 $\pm$ NR	T: 112 I: 34 $C_{1,2}$: 33,45	VRET + CBT	C_1: Waitlist C_2: CBT	Anxiety symptoms (LSAS)	21.4%	N/N	N/N/Y
Wallach et al., 2011 (Israel)	2-arm RCT	Public speaking anxiety (self-diagnosis)	28 $\pm$ NR	T: 20 I: 10 C_1: 10	VRET + CT	C_1: CT	Anxiety symptoms (LSAS)	0%	N/N	N/N/Y
Zainal et al., 2021 (USA)	2-arm RCT	Social anxiety disorder (DSM-5)	23.3 $\pm$ 9.32	T: 44 I: 26 C_1: 18	VRET	C_1: Waitlist	Anxiety symptoms (SIAS)	20.45%	Y/Y	N/N/Y

Note: 2D = Two dimensions; 3D = Three dimensions; ABM = Attention Bias Modification in Virtual Reality; C_1 = Comparator with control condition; C_2 = Comparator with other intervention; CBT = Cognitive Behavioral Therapy; CT = Cognitive therapy; DSM-4-TR = Diagnostic and Statistical Manual of Mental Disorders (4th, ed, text, rev); DSM-5 = Diagnostic and Statistical Manual of Mental Disorders (5th ed); EX*P* = Exposure program; FNE-B = Fear of Negative Evaluation – Brief Form; I = Intervention; In-vivo ET = In vivo Exposure Therapy; ITT, Intention-to-treat analysis; LSAS = Liebowitz Social Anxiety Scale; LSAS-SR = Liebowitz Social Anxiety Scale Self-Reported version; M = Mean; MDM = Missing data management; N = No; NR = Not reported; NA = No Additional Intervention; PE = Psychoeducational program; PRCS = Personal Report of Confidence as a Speaker inventory; PSAS = Public Speaking Anxiety Scale; RCT = Randomized Controlled Trial; SD = Standard Deviation; SIAS = Social Interaction Anxiety Scale, T = Total; USA: United States of America; VR-D = Virtual Reality Distraction; VRET = Virtual Reality Exposure Therapy; Y = Yes.

et al., 2009, 2011). The duration lasts from 1 d (Lindner et al., 2019) to 14 weeks (Bouchard et al., 2017), and follow-up ranges from 3 months (Burton et al., 2013) to 6 years (Anderson et al., 2017).

Risk of bias version 2 assessment

Figure S1 presents the summary of bias with a satisfactory inter-rater agreement ($k = 0.88$). Five trials used ITT analysis, and twelve trials used per-protocol analysis. Most trials with ITT analysis (56%) and per-protocol analysis (48.3%) found a low risk of bias ratings across five domains. Among five trials with ITT analysis, participants were not blind to the condition before allocation in Lindner et al. (2019) thus they rated it as high risk. Eight trials (47%) did not provide information on allocation concealment. Attrition rates range from 0% (Ma et al., 2020) to 27% (Kahlon, Gjestad, et al., 2023; Kahlon, Lindner, et al., 2023). There were some concerns that the result was biased by missing outcome data in twelve trials (70%). Participants who serve as outcome assessors were aware of the intervention received in all trials except two trials (Anderson et al., 2013; Kampmann, Emmelkamp, et al., 2016). Eight trials published protocol or were registered in various clinical registries and were unclear for reporting bias in more than half trials (52.9%).

VRET compared with waitlist comparators

A meta-analysis pooling the results of 11 RCTs on 485 participants is conducted to evaluate the effects of VRET compared with the waitlist on anxiety symptoms, as shown in Figure 2. At the post-intervention, significant differences exist between VRET and waitlist comparators ($t = 4.25$, $p = 0.001$) with small to medium effect size ($g = 0.48$; 95% CI = 0.23, 0.73). At the follow-up at 3 months, two arms in one RCT are pooled among 47 participants, and a medium to large effect ($g = 0.76$; 95% CI = 0.45, 1.07) is found. A significant difference existed between the two groups ($t = 31.38$, $p = 0.02$). Moderate heterogeneity is detected post-intervention ($I^2 = 34\%$). These results

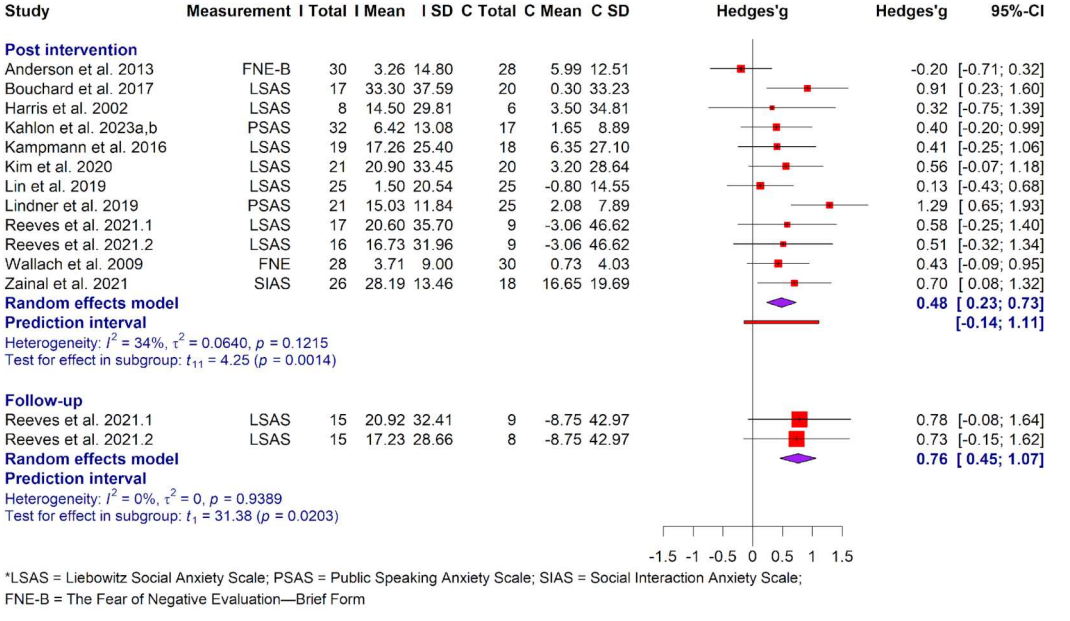


Figure 2. Forest plot of effect size (Hedges' *g*) of pre-post changes and pre-follow-up changes on anxiety symptoms for virtual reality exposure therapy and waitlist comparators at post intervention and follow-up.

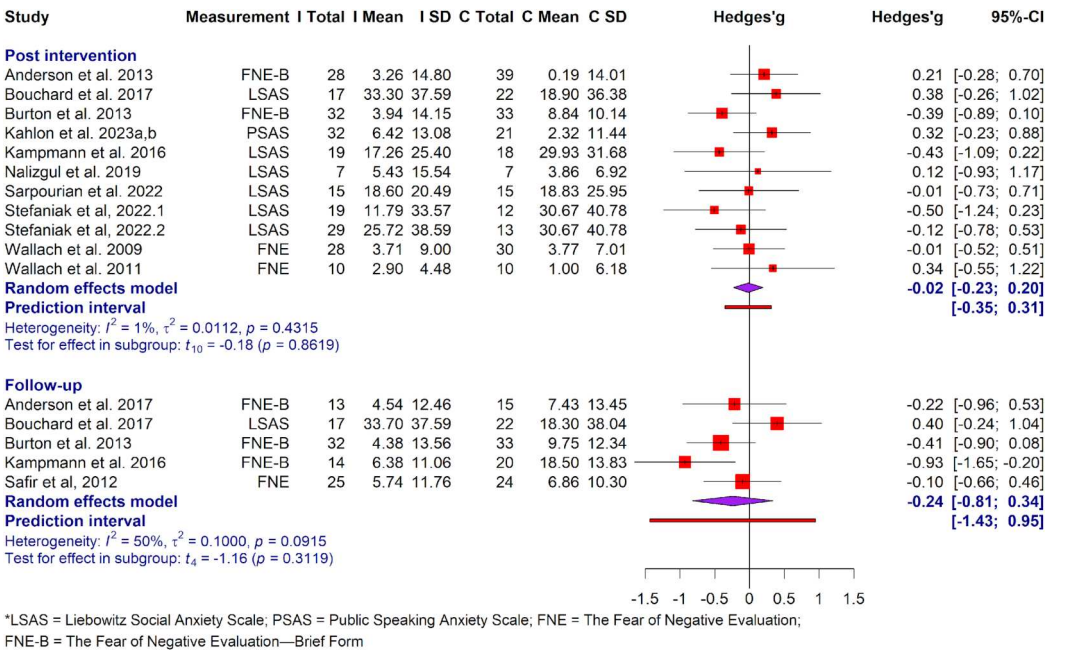


Figure 3. Forest plot of effect size (Hedges' *g*) of pre-post changes and pre-follow-up changes on anxiety symptoms for virtual reality exposure therapy and other interventions at post intervention and follow-up.

confirm our first hypothesis in this review that VRET would demonstrate a greater effect compared with the control condition on anxiety symptoms at post-intervention and follow-up assessment.

VRET compared with other interventions

Ten RCTs contribute to the evaluation of VRET with other interventions on anxiety symptoms among 456 participants post-intervention (Figure 3). In the meta-analysis, the other intervention includes exposure in vivo, CBT, psychoeducation, and counseling. The meta-analysis reveals no significant differences between VRET and other interventions ($t = -0.18$, $p = 0.86$). Five RCTs involving 215 participants are pooled at follow-up from 3 months to 6 years. Similarly, no significant differences are detected ($t = -1.16$, $p = 0.31$). Moderate heterogeneity is found at follow-up ($I^2 = 50\%$). These findings confirm our second hypothesis that VRET would demonstrate a similar effect when compared to other interventions.

Subgroup analyses and meta-regression analyses

A series of subgroup analyses are conducted by region, population, and co-intervention for two comparisons, as shown in Table 2. There are significant differences between subgroups when co-intervention is present for VRET ($\chi^2 = 1.76$, $p < 0.01$) compared to the waitlist comparators. Trials with co-intervention have the greatest effect ($g = 0.73$; 95% CI = 0.05, 1.41) when compared to trials without co-intervention ($g = 0.33$; 95% CI = 0.05, 0.60). However, VRET shows no subgroup differences by various regions or nature of population type compared to the waitlist comparators. Subgroups show significant group differences when different natures of the population are found ($\chi^2 = 4.35$, $p = 0.037$) for VRET compared with the other interventions. The population with symptomatic SAD showed a significantly larger effect size ($g = 0.21$; 95% CI = −0.06, 0.49) when compared to the population with diagnostic SAD ($g = -0.10$; 95% CI = −0.41, 0.20). However, VRET does not show any differences in subgroups by various regions, or the presence of co-intervention compared to the other interventions.

Table 2. Subgroup analyses for virtual reality exposure therapy compared with different comparators at post-intervention.

Design	Subgroups	No. of studies	No. of arms	Sample size	Heterogeneity (I^2)	Effect size (Hedges' *g*) (95% CI)	Overall effect *t*, *p*	*Chi-square* (^{2}) subgroup differences
VRET versus waitlist comparators								
Region	North America	4	4	153	63%	0.42 (−0.41, 1.24)	$t = 1.61$, $p = 0.206$	$^2 = 1.76$,
	Europe	4	5	183	23%	0.64 (0.16, 1.13)	$t = 3.66$, $p = 0.022$	$p = 0.414$
	Asia	3	3	149	0%	0.36 (−0.17, 0.89)	$t = 2.90$, $p = 0.101$	
Population	Diagnostic	6	6	275	41%	0.43 (0.03, 0.84)	$t = 2.76$, $p = 0.040$	$^2 = 0.20$,
	Symptomatic	5	6	210	35%	0.54 (0.09, 0.99)	$t = 3.08$, $p = 0.028$	$p = 0.651$
Co-intervention	Yes	4	4	190	47%	0.73 (0.05, 1.41)	$t = 3.43$, $p = 0.041$	$^2 = 1.76$,
	No	7	8	295	0%	0.33 (0.05, 0.60)	$t = 2.83$, $p = 0.026$	$p = 0.098$*
VRET versus other interventions								
Region	North America	3	3	171	56%	0.04 (−0.97, 1.06)	$t = 0.18$, $p = 0.874$	$^2 = 0.78$,
	Europe	4	5	177	13%	−0.11 (−0.57, 0.35)	$t = -0.68$, $p = 0.535$	$p = 0.676$
	Asia	3	3	108	0%	0.05 (−0.35, 0.46)	$t = 0.58$, $p = 0.619$	
Population	Diagnostic	6	7	339	19%	−0.10 (−0.41, 0.20)	$t = -0.82$, $p = 0.444$	$^2 = 4.35$,
	Symptomatic	4	4	117	0%	0.21 (−0.06, 0.49)	$t = 2.48$, $p = 0.089$	$p = 0.037$*
Co-intervention	Yes	6	7	308	21%	−0.02 (−0.34, 0.31)	$t = -0.13$, $p = 0.898$	$^2 = 0.00$,
	No	4	4	148	0%	−0.01 (−0.49, 0.47)	$t = -0.05$, $p = 0.964$	$p = 0.959$

Note: CI = Confidence interval; I^2 = Heterogeneity; No. = Number; *t* = t-statistics; VRET = Virtual Reality Exposure Therapy; *$p < 0.1$

A series of univariate random-effects meta-regression analyses is conducted to examine the effect of covariates on treatment effect for two comparisons (Table 3). We found that the publication year ($\beta = 0.01$–0.02, $p = 0.33$–0.68), number of sessions ($\beta = -0.01$–0.60, $p = 0.60$–0.84), attrition rate ($\beta = -0.02$– < 0.01, $p = 0.35$–0.75), and sample size ($\beta = 0.01$– < 0.001, $p = 0.20$–0.95) did not have a significant effect on anxiety symptoms when we compared VRET to waitlist or other interventions. We found no significant covariates for the two comparisons.

Narrative synthesis

We find one trial (Moldovan & David, 2014) that does not provide sufficient information for our meta-analysis, and emails are sent to authors for additional information without responses. Accordingly, narrative synthesis is performed for this trial. The study shows no significant differences between one session of VRET plus CBT compared with the waitlist group among 32 participants with social and specific phobias (Moldovan & David, 2014). However, a pre–post result shows that participants experience less social phobia (t (13) = 5.48, $p < 0.05$) after the intervention.

Certainty of evidence

The certainty of evidence for all outcomes is reported according to GRADE criteria (Table S7). The overall evidence ranges from very low to moderate. The high percentages of high or unclear risk of biases and the presence of heterogeneity are reasons for downgrading. There was no evidence of publication bias through visual inspection of asymmetrical funnel plots (Figures S2(A) and (B)) and Egger's tests ($p = 0.29$–0.91) for two comparisons.

Discussion

Summary of findings

A total of 17 RCTs in 20 publications involving 943 participants with SAD across 12 countries are included in this review. Our results support the first hypothesis that VRET would have greater efficacy than the waitlist comparators in reducing anxiety symptoms at post-intervention and follow-up assessment. The meta-analyses support our second hypothesis that VRET would demonstrate a similar effect compared with other interventions at post-intervention and follow-up. Comparison of VRET with the waitlist comparators through subgroup analyses reveals that VRET plus co-intervention is more effective than solely VRET in reducing anxiety symptoms. When comparing VRET to other interventions, subgroup analysis showed that participants with symptomatic SAD demonstrated a greater effect on reducing anxiety symptoms than those with diagnostic SAD.

Short- and long-term effects of VRET versus waitlist comparators

Our review reveals that the superiority of VRET over the waitlist comparators at post-intervention is consistent with the findings of previous reviews (Carl et al., 2019; Chesham et al., 2018). The corrective element of emotional process theory may help participants by modifying the fear structure (Foa et al., 2006). VR technology provides virtual exposure to a fear to stimulate that allows the creation of safe and controlled environments, personalized to individual needs (Carvalho et al., 2010). VRET can easily generate corrective information to target erroneous thoughts and disconfirm presuppositions present in the pathological fear structure (McNally, 2007; Wilhelm et al., 2005). This in turn can reduce anxiety symptoms for SAD compared with the control condition.

Our result indicates that effectiveness can be maintained at follow-up. However, it is too early to conclude the long-term effect of VRET because our review involves only two arms of one trial.

Table 3. Random-effects meta-regression analyses by potential covariates.

	Univariate meta-regression					
Covariate	*B*	SE	95% Lower	95% Upper	*t*	*p*
VRET versus waitlist comparators						
Year of publication	0.02	0.02	−0.03	0.07	1.03	0.33
Number of sessions	−0.01	0.03	−0.07	0.06	−0.20	0.84
Attrition rate	−0.02	0.02	−0.06	0.02	−0.98	0.35
Sample size	0.01	0.00	−0.02	0.00	−1.37	0.20
VRET versus other interventions						
Year of publication	0.01	0.02	−0.04	0.06	0.43	0.68
Number of sessions	0.60	0.02	−0.07	0.04	−0.54	0.60
Attrition rate	<0.001	0.01	−0.03	0.02	−0.33	0.75
Sample size	<0.001	<0.001	−0.01	0.01	0.06	0.95

Note: VRET = Virtual Reality Exposure Therapy; β = Standardized regression coefficients (Beta); *SE* = Standard Error; *t* = *t* statistics.

Therefore, additional research is needed to further examine the long-term efficacy of VRET compared with the waitlist comparators.

Short- and long-term effects of VRET versus other interventions

Equivalent effects are found for VRET and other standard interventions at post-intervention and follow-up evaluations. The other standard intervention includes exposure in vivo, CBT, psychoeducation, or counseling in the current review. Our results are in line with previous reviews (Horigome et al., 2020; Kampmann et al., 2016b) claiming an equivalent effect between VRET and other interventions. However, the findings are inconsistent with the two reviews (Powers & Emmelkamp, 2008; Wechsler et al., 2019). These discrepancies can be explained by the possibility of a certain variance between participants, as well as the different number of included trials, types of research design, and regime of the VRET or VR environment. Equivalency of VRET and other standard interventions is found, possibly because VRET contains common features of exposure-based behavioral intervention (Pittig et al., 2019). Another possibility is the technical development of VR devices (Park et al., 2019) that could have enhanced the therapy results in VRET. Such similar ingredients and enhancements of VR technology can explain the similar treatment effects of VRET and other interventions in improving anxiety symptoms.

At follow-up ranging from 3 months to 6 years, anxiety symptom scores are similar between VRET and other interventions, suggesting the long-term benefits of VRET. This result differs from a previous review (Horigome et al., 2020). Our results showed that VRET is more effective than in vivo exposure at 6 months, but in vivo exposure is more effective than VRET at >12 months. The difference may be linked to the different research design, year of publication, and data analysis methods in both reviews. Although more trials (17 RCTs) are included in the current review than in the previous investigation (11 RCTs and 11 non-RCTs) (Horigome et al., 2020), the analysis still relies on a small number of studies. Future investigations are warranted. As VR materials and procedures continuously improve, ideal environments are important to create for exposures and appropriate VR scenario development to maximize the effectiveness of VRET and treatment compliance.

Co-intervention and participants with symptomatic SAD

When we compared VRET with waitlist comparators, we observed that VRET plus co-interventions such as CBT had a larger effect size than VRET alone. CBT is regarded as a first-line treatment for SAD that aims at altering maladaptive behavior and cognition by administering behavioral and cognitive strategies (Pelissolo et al., 2019). This finding is consistent with a previous one (Wechsler et al., 2019), suggesting that a combination of VR exposure with cognitive intervention may be

advantageous in treating social phobia owing to the realization that virtual interactions target central fears. Hence, whether applied solely as VRET or in combination with CBT, our review offers new treatment possibilities.

We further observed a greater effect in individuals with self-reported symptoms of SAD than those diagnosed with SAD when we compared the VRET with other interventions. This finding seems to be a possible explanation for different treatment-seeking behaviors in both groups (Calear et al., 2021). Individuals with diagnostic SAD may not experience the social anxiety symptoms in all situations, and people with the condition may delay seeking treatment, resulting in differences in treatment responses (Koyuncu et al., 2019; McDonagh et al., 2022). However, evidence related to levels of symptoms of social anxiety and treatment-seeking behaviors is limited. Further research should address this issue as well.

Strengths and limitations

The strengths of this meta-analysis are that all studies included are RCTs. The three-step search strategy is performed from nine databases and two main clinical registries, thereby ensuring a comprehensive search (Aromataris & Munn, 2020). The present results add prior meta-analyses of the effectiveness of VRET for SAD by (1) including a sizeable total number of RCTs (10-11 RCTs for each comparison) in a total of 17 RCTs published in 20 articles across 12 countries, (2) reporting certainty of the evidence of all outcomes, (3) assessing the long-term effect of VRET in two comparisons, and (4) identifying greater effects among individuals with symptomatic SADs if VRET is combined with CBT.

However, this review has some limitations. First, variations exist in the treatment procedure, e.g., the VR system and equipment in each RCT can influence the efficacy of VRET. Second, the selected RCTs use self-reported questionnaires to assess anxiety symptoms that could have social desirability biases. Third, the long-term effect of VRET compared with the waitlist is based solely on two arms of 1 trial which limit the generalizability of the result. Fourth, the other interventions included various interventions (in-vivoET, online psychoeducation, CBT-based psychoeducation, or counseling) that might not provide the same effects. Finally, few RCTs have sufficient study quality, and overall evidence ranges from very low to moderate, which may decrease the internal validity of the findings.

Clinical implications

VRET can provide a supplementary therapy for improving anxiety symptoms. VRET may be more likely to reach individuals with SAD who may be too afraid to engage in face-to-face contact with a therapist. VRET has some advantages because it overcomes some limitations of the vivo technique (de Carvalho et al., 2010). VRET is also found to be significantly more practical for therapists than in vivo exposure (Pelissolo et al., 2019). Our review suggests that VRET plus CBT can lead to a better treatment effect than VRET alone. Considering that VR systems have drastically changed over the last decade, therapists or users require special training to operate or create VRETs. As VRET is time effective, accounting for fewer organizational effects, achieving efficacious exposure treatment for a wider population of patients has become more possible. VRET should be considered a useful, efficacious, and potential therapeutic tool (Carl et al., 2019). Additional rigorous long-term investigations are needed before VRET becomes a routine clinical practice.

Future studies

According to GRADE criteria showing that the overall evidence ranges from low to moderate, additional high-quality and large-scale trials with long-term follow-up are needed based on the recommendations of the CONSORT statement (Schulz et al., 2010). Future trials should perform allocation concealment and blinding of participants and assessors to minimize biases (Drucker et al., 2016; Laher, 2016). Variation exists among VRET content, number of sessions, length of each

session, frequency, and duration, so further investigation may uncover the ideal way to utilize VRET. To illustrate, it will be noteworthy to investigate the effects of VRET in combination with another evidence-based treatment (e.g., CBT) in comparison with VRET alone. Given that the quality of the VR technology used can influence the treatment effect, further investigation is necessary to identify the most appropriate virtual environment, VR equipment, movement mode in VR, and stimulus for treating SAD. As VRET is performed mostly with therapist involvement, more research should be conducted on VRET administered without the presence of a therapist to develop a self-directed rendition of VRET (Lindner et al., 2019) that can be performed within the home setting (Oing & Prescott, 2018). Consequently, treatment accessibility can be enhanced as technology continues to improve, allowing VRET to be received using mobile applications and affordable HMDs to treat themselves at their own pace.

Conclusion

In reducing anxiety symptoms, VRET tends to be more efficacious than the waitlist comparators and equally efficacious to other interventions at post-intervention and follow-up. Due to the small number of trials at follow-up assessment, more work is needed to ascertain the long-term effect of VRET among individuals with SAD. Nevertheless, VRET can be considered an alternative to the standard intervention for individuals with SAD. VRET appears to be beneficial if combined with CBT targeting individuals with symptomatic SAD. Given that the quality of existing RCTs is far from ideal, more rigorous RCTs are required to confirm VRET efficacy in two comparisons, especially long-term effects. Future studies are warranted to investigate how to maximize the treatment efficacy of VRET in an individual with SAD.

Acknowledgements

The reviewers thank the authors of the included studies for providing us with additional information

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contribution

YLT: Conceptualization; investigation; data curation; writing – original draft. VC: Investigation; data curation; writing – review and editing. DA: Conceptualization; formal analysis; software; visualization; investigation; data curation; writing – original draft; writing – review and editing. WWA: Formal analysis; software; visualization; data curation; writing – original draft; writing – review and editing. YL: Conceptualization; formal analysis; software; visualization; investigation; data curation; writing – original draft; writing – review and editing; supervision.

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